

Brian Pulfer

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EDUCATION

UNIGE - University of Geneva

Ph.D. in Machine Learning

Geneva, Switzerland

Nov. 2021 – Sep. 2026

USI - University of Southern Switzerland

MSc in Artificial Intelligence (GPA 9.1/10)

Lugano, Switzerland

Aug. 2019 – Jun 2021

SUPSI - University of Applied Sciences of Southern Switzerland

BSc in Computer Science (GPA 5.0/6)

Manno, Switzerland

Aug. 2016 – Jun 2019

EXPERIENCE

Ph.D. Software Engineering Intern

Meta Platforms Inc

June 2025 – September 2025

Menlo Park, California, United States of America

- NVFP4 quantization of Large Language Models run on the vLLM inference engine
- Achieved memory reduction of **67.5%/37.6%** (w.r.t BF16/FP8) with minimal accuracy drop ($< 2\%$).
- Exploratory studies of NVFP4 quantization of KV-cache in vLLM

Ph.D. Student in Machine Learning

University of Geneva

November 2021 – November 2026

Geneva, Switzerland

- Researching SSL vision models and relative robustness to adversarial attacks.
- Teaching Assistant for Algorithms and Data Structures courses.

Machine Learning Intern

University of Southern Switzerland

July 2020 – August 2020

Lugano, Switzerland

- Developed a tool for automatic collection and cleaning of a dataset through web crawling and heuristics.
- Applied state-of-the-art techniques for image classification, segmentation, and object detection.

Competitions & Hackathons

- Classified 3rd in the *NeurIPS Weather4cast Challenge 2022* by leveraging transformers for videos.
- USI Hackathon 2019, START Hack 2021, HackZürich 2022, Winner of HackZürich 2023.

SELECTED PUBLICATIONS

- *Task-Agnostic Attacks Against Vision Foundation Models*
The 5th Workshop of Adversarial Machine Learning on Computer Vision: Foundation Models + X
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025)
- *Robustness Tokens: Towards Adversarial Robustness of Transformers*
18th European Conference on Computer Vision ECCV 2024
- *Mind the Gap! A Study on the Transferability of Virtual vs Physical-world Testing of Autonomous Driving Systems.*
IEEE Transactions on Software Engineering

LEADERSHIP & AWARDS

Hackathon Organizer

Nov. 2020 - Nov 2021

- Organizer of the first edition of the *Formula USI* Hackathon at the University of Southern Switzerland.

SODESKA Scholarship

April 2021

- Scholarship awarded on merit to students of the University of Southern Switzerland.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, CUDA, JavaScript, Java, SQL

Frameworks: Pytorch, Jax, Flax, Tinygrad

Developer Tools: Git, CI/CD, Linux, ssh, SLURM, pip, uv, Docker, VSCode, Copilot, Claude code, codex

Libraries: accelerate, alumentations, datasets, diffusers, einops, scikit-learn, opencv-python, hydra, numpy, matplotlib, openai pandas, pre-commit, seaborn, timm, transformers, trl, torchvision, vllm, wandb, xformers